INCH-POUND

MIL-F-70807 (AR)
AMENDMENT 6
20 February 1992
SUPERSEDING
AMENDMENT 5
19 September 1991

MILITARY SPECIFICATION

FUZE, PD: M745
(LESS BOOSTER AND LEAD)

This amendment forms a part of Military Specification MIL-F-70807 (AR), dated 11 March 1988, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Department and Agencies of the Department of Defense.

PAGE 2

2.1.2, DRAWINGS, Add "9347415 - Packing & Marking of Box Fiberboard for Fuze, Multi-Option, M734 and Fuze, PD, M745" and "9381685 -. Packing & Marking of Container, M548 for Fuze, Multi-Option, M734 and Fuze, PD, M745."

Delete old drawing number and substitute new drawing number wherever it appears in this specification as follows:

| Old Dwg. No. | New Dwg. No. |
|---|---|
| 11737001 | 11927103 |
| 11737000 | 11927102 |
| 11737007 | 11927120 |
| 11744131 | 11927094 |
| 11740827 | 11927066 |
| 11740828 | 11927087 |
| 11740824 | 11927137 |
| 11744134 | 11927095 |
| 11744135 | 11927116 |
| 11744136 | 11927117 |
| 11744145 | 11927118 |
| 11737030 | 11927109 |
| 11737031 | 11927125 |
| 11737029 | 11927108 |
| 11744139 | 11927141 |
| 11744140 | 11927142 |
| 11744133 | 11927140 |
| 11744152 | 11927146 |
| 11737024 | 11927124 |



AMSC N/A FSC 1390

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PAGE 2

| Old Dwg. No. | New Dwg. No. |
|---|---|
| 11740823 | 11927136 |
| 11744129 | 11927093 |
| 11744128 | 11927092 |
| 11744147 | 11927144 |
| 11744146 | 11927143 |
| 11744126 | 11927139 |
| 11737016 | 11927122 |
| 11740821 | 11927084 |
| 11744127 | 11927091 |
| 11737012 | 11927106 |
| 11744137 | 11927096 |
| 11744138 | 11927097 |
| 11744130 | 11927115 |
| 11744141 | 11927098 |
| 11737020 | 11927079 |
| 11737021 | 11927080 |
| 11737027 | 11927081 |
| 11740819 | 11927134 |
| 11744125 | 11927138 |
| 11737013 | 11927121 |
| 11740806 | 11927130 |
| 11744121 | 11927067 |
| 11737010 | 11927105 |
| 11737026 | 11927107 |
| 11737016 | 11927122 |
| 11744148 | 11927074 |
| 11737023 | 11927123 |
| 11744151 | 11927145 |
| 11737000 | 11927102 |
| 11740826 | 11927086 |
| 11740828 | 11927087 |
| 11740829 | 11927088 |
| 11744120 | 11927090 |
| 11737030 | 11927109 |
| 11737001 | 11927103 |

PAGE 3

3.1.1: Change "458" to read "436", and change "428" to read "406".

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PAGE 4

3.1.4: Delete in its entirety and substitute the following:

"3.1.4 Ballistic sample.

3.1.4.1 Initial production lot. Select at random ballistic lot acceptance samples from first production lot. Ballistic lot acceptance test plan is shown in figure 2.

3.1.4.2 Production lot. Select at random ballistic lot acceptance samples from each production lot. Ballistic lot acceptance test plan is shown in figure 3."

3.2: Change "(Figure 3)" to read "(Figure 4)."

PAGE 5

3.3: Change "-25 degrees F" to read "-50 degrees F."

* 3.3.1 Change first sentence from "The fuze turbine shall not rotate more than one revolution due to air flow into the nozzle prior to actual gun firing".

To: "The fuze turbine shall not rotate more than three revolutions due to air flow into the nozzle prior to actual gun firing".

PAGE 6

* 3.3.2.1 Change first sentence from "With the zigzag body in the forward position, the spring gear latch shall prevent rotation of more than one revolution of the turbine under .206+.000, -.006 psig air pressure" to read "With the zig-zag body in the forward position, the spring gear latch shall prevent rotation of more than three revolution of the turbine under .206+.000, -.006 psig air pressure".

PAGE 7

3.3.2.4: Line 4, Change "low" to read "low (-50 degrees F)".

* 3.3.2.5: Delete in its entirety and substitute as follows:



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"3.3.2.5 Spring, gear latch (destructive, non-arming). With the zig-zag body in the forward (up) position, the spring, gear latch shall prevent rotation of the gear train with 0.70 ± 0.05 psig air pressure applied to the turbine for at least 3 seconds. The S&As shall have passed this test if:

a. The spring, gear latch prevented rotation of no more than 10 revolutions of the turbine, or

b. The jackscrew setting does not become more than .659 inch. This dimension applies with the zig-zag locked down and the rotor assembly in the fully seated (toward the barrier) position.

The S&A device need not be operable but shall be in the safe position following this test."

PAGE 9

4.4.1. Delete in its entirety and substitute the following:

"d. The lot size shall be a maximum of 50,000 fuzes or one month's production, whichever is smaller. Minimum lot size shall be 1,000 fuzes for lots 1 and 2 and 5,000 fuzes thereafter."

PAGE 10

4.4.3.1: Change "Figure 2" to read Figure 2 and 3."

4.4.3.2: Change "Figure 2" to read "Figure 2 and 3."

4.4.3.2: Delete the last sentence of the paragraph.

PAGE 13

4.5.1.1.2.1: Delete the words "high explosive of the."

PAGE 15

4.5.1.2.1.5: Change "Test 101.3, Procedure I" to read "Test 101.2, Procedure II."

4.5.1.2.1.6: Change "Test 102.2" to read "Test 102.1."

4.5.1.2.1.7: Change "Test 103.2" to read "Test 103.1."

* 4.5.1.3 Delete the second sentence, "Observance of any turbine rotation shall be made visually".

4.5.2.1: Change "The S&A Device shall be loaded with a conductive dummy detonator and placed in the tester of 4.4.4.1.2." to "The S&A Device shall be placed in the tester of 4.4.4.1.2."

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PAGE 16

* 4.5.2.5 Delete in its entirety and substitute as follows:

"4.5.2.5 Spring, gear latch (destructive, non-arming test). The S&A Device shall be placed in the tester of 4.4.4.1.2 with the zig-zag body in the forward position, apply an air pressure of 0.70 ± 0.05 psig. The requirements of 3.3.2.5 shall be met."

PAGE 17

6.2.2: Delete last sentence of paragraph.

PAGE 21

Table I: Delete "11740827 Pad, Booster 10"
"11740828 Cap, Booster 10"

PAGE 26

4.4.2.3: Major 101, change "Missing stab primer spring ...0.40% ....3.1...Probe/IG test" to read "Missing stab primer spring ...100%...3.1...Probe/IG test."

PAGE 27

4.4.2.4: Major 102, Change ".040" to read ".042."

PAGE 31

4.4.2.8, Major 101, delete in its entirety

Add Major 103, 104 as follows:

"103, .027 length out of tolerance....0.40%..3.1..Gage

104, Firing pin not perpendicular to datum D within 0.003 at max. material condition ....0.40%...3.1...Gage."

PAGE 33

4.4.2.10, Add Major 104 as follows:

"104, Improper perpendicularity of pin diameter...0.40%.... 3.1...Gage."

PAGE 34

4.4.2.11: Minor 201, Change ".264" to read ".268."



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PAGE 36

4.4.2.13: Change Note A:

From: "Two-thirds of the sample shall be conditioned to -50 degrees F and one-third to +145 degrees F for two hours minimum. The barrier shall be assembled into a loaded S&A (11737010). The S&A explosives shall be initiated electrically with the rotor in the safe position. There shall be no evidence of fragmentation, cracking or breakthrough of the barrier, bulging in excess of .015 inch on booster side face, or scorching of lead charge explosives after the M61 detonator has fire."

To: "Two-thirds of the sample shall be conditioned to -50 degrees F and one-third to +145 degrees F for two hours minimum. The barrier shall be assembled into a loaded S&A (11737010). The S&A explosives shall be initiated mechanically by stabbing the delay primer with the rotor in the safe position. There shall be no evidence of fragmentation, cracking or break through of the barrier, bulging in excess of .015 inch on booster side face, or scorching of lead charge explosives after the M61 detonator has fired."

Note A, Add: "Special inspection level S-4 shall be applied for selecting sample size."

PAGE 46

4.4.2.22: Delete paragraph in its entirety and substitute new paragraph included.

PAGE 55

4.4.2.31: Delete Major 101 in its entirety and substitute "None defined."

PAGE 64

Delete in its entirety and substitute included pages 64 and 64a.

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PAGE 64a

Change Duds "3/4" to "6/7".

PAGE 65

Delete in its entirety and substitute per included sheet.

PAGE 66

Delete in its entirety and substitute per included sheet.

PAGE 67

Figure 4: Add new figure per included sheet.



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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

MIL-F-70807 (AR)
AMENDMENT 6

| PARAGRAPH | TITLE | SHEET OF | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.22 | CAM | | 11744137 |
| | | | NEXT HIGHER ASSEMBLY 11737016 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Insufficient strength of arm | | 0.40% | 3.1 | Force test (Dwg. Note 5) |
| 102 | .203 arm height or perpendicularity out-of-tolerance | | 0.40% | 3.1 | Gage |
| 103 | .140 arm length out-of-tolerance | | 0.40% | 3.1 | Gage |
| Minor | | | | | |
| 201 | .125 diameter out-of-tolerance | | 0.65% | 3.1 | Gage |
| 202 | .500 length out-of-tolerance | | 0.65% | 3.1 | Gage |
| 203 | Evidence of poor workmanship | | 0.65% | 3.5 | Visual |

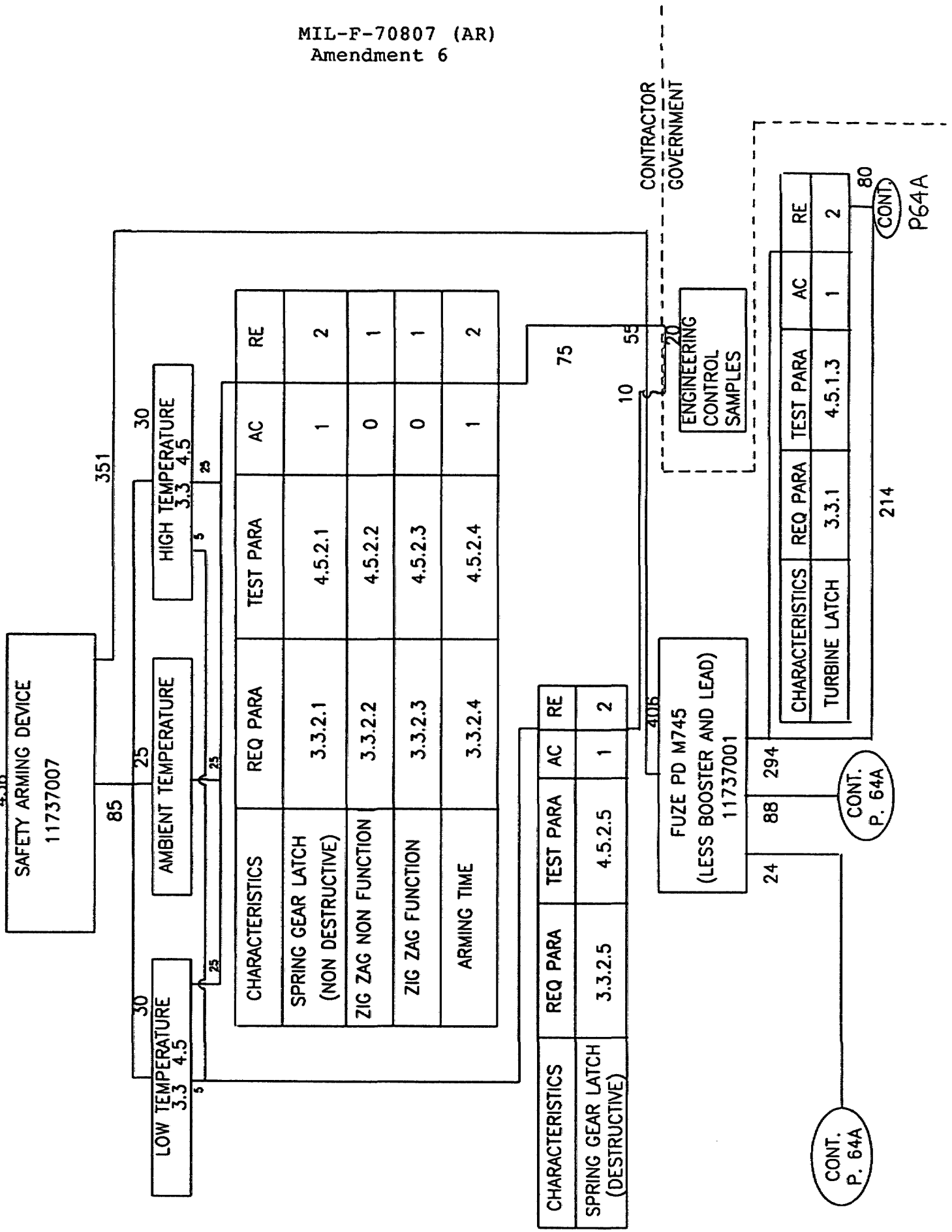
NOTES:



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FIGURE 1

FIRST ARTICLE INSPECTION FLOW CHART




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FIGURE 1 (Cont'd)

CONT.

CONT.

CONT.

FUZE M745
(WITH INERT BOOSTER AND LEAD)

| ENVIRONMENTAL REQ. | REQ. PARA. | TEST PARA. | N | SAFE AC | SAFE RE | OPER AC | OPER RE | |
|---|---|---|---|---|---|---|---|---|
| TEMP. AND HUMIDITY | 3.3.1.3.1 | 4.5.1.2.1.1 | 14 | 0 | 1 | 2 | 3 | 14 |
| SALT FOG | 3.3.1.3.1 | 4.5.1.2.1.2 | 8 | 0 | 1 | 0 | 1 | 4 |
| DROP TEST ( 1.5 M) | 3.3.1.3.1 | 4.5.1.2.1.3 | 20 | 0 | 1 | 2 | 3 | 10 |
| T.V. TEMPERATURE | 3.3.1.3.1 | 4.5.1.2.1.4 | 16 | 0 | 1 | 2 | 3 | 16 |
| JOLT | 3.3.1.3.2 | 4.5.1.2.1.5 | 12 | 0 | 1 | NA | NA | |
| JUMBLE | 3.3.1.3.2 | 4.5.1.2.1.6 | 8 | 0 | 1 | NA | NA | |
| 40 FT. DROP | 3.3.1.3.2 | 4.5.1.2.1.7 | 10 | 0 | 1 | NA | NA | |

TO GROUP C BALLISTICS TEST PLAN

44

294

FUZE PD, M745
LOADED 11737000

44

BALLISTIC TEST PLAN
REQ. PARA. 3.3.1.1 & 3.3.1.2
TEST PARA. 4.5.1.1
N=338

| GROUP | CHG. | QE | +145F | +70F | -25F | -50F | TOTAL |
|---|---|---|---|---|---|---|---|
| A ** | 0 | 85 | 25 | 25 | 25 | 25 | 100 |
| B | 4 | 45 | 25 | 163 | – | – | 188 |
| C * | 4 | 45 | 25 | – | 25 | – | 50 |
| TOTAL | | | 75 | 188 | 50 | 25 | 338 |

ACCEPT/REJECT CRITERIA
1. BY DEFECT CATEGORY

| | AC/RE |
|---|---|
| DUDS | 6/7 |
| PREMATURES | 0/1*** |

ENGINEERING CONTROL SAMPLES

*Rounds from drop test (1.5M) that fail to function will not count towards the accept/reject dud criteria.

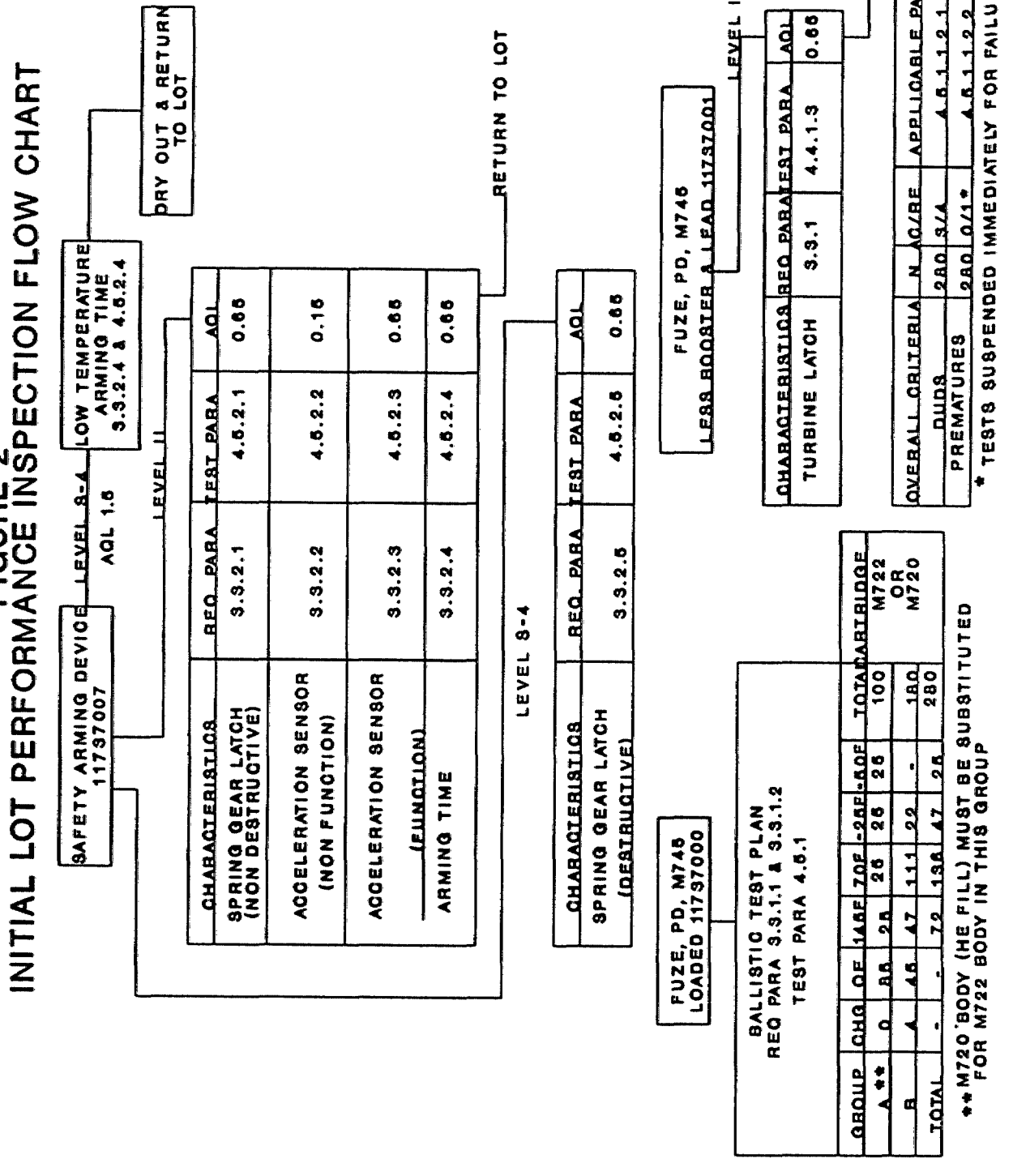
**M720 body (H.E. Fill) must be substituted for M722 body in this group.

***Test suspended immediately for failure analysis.



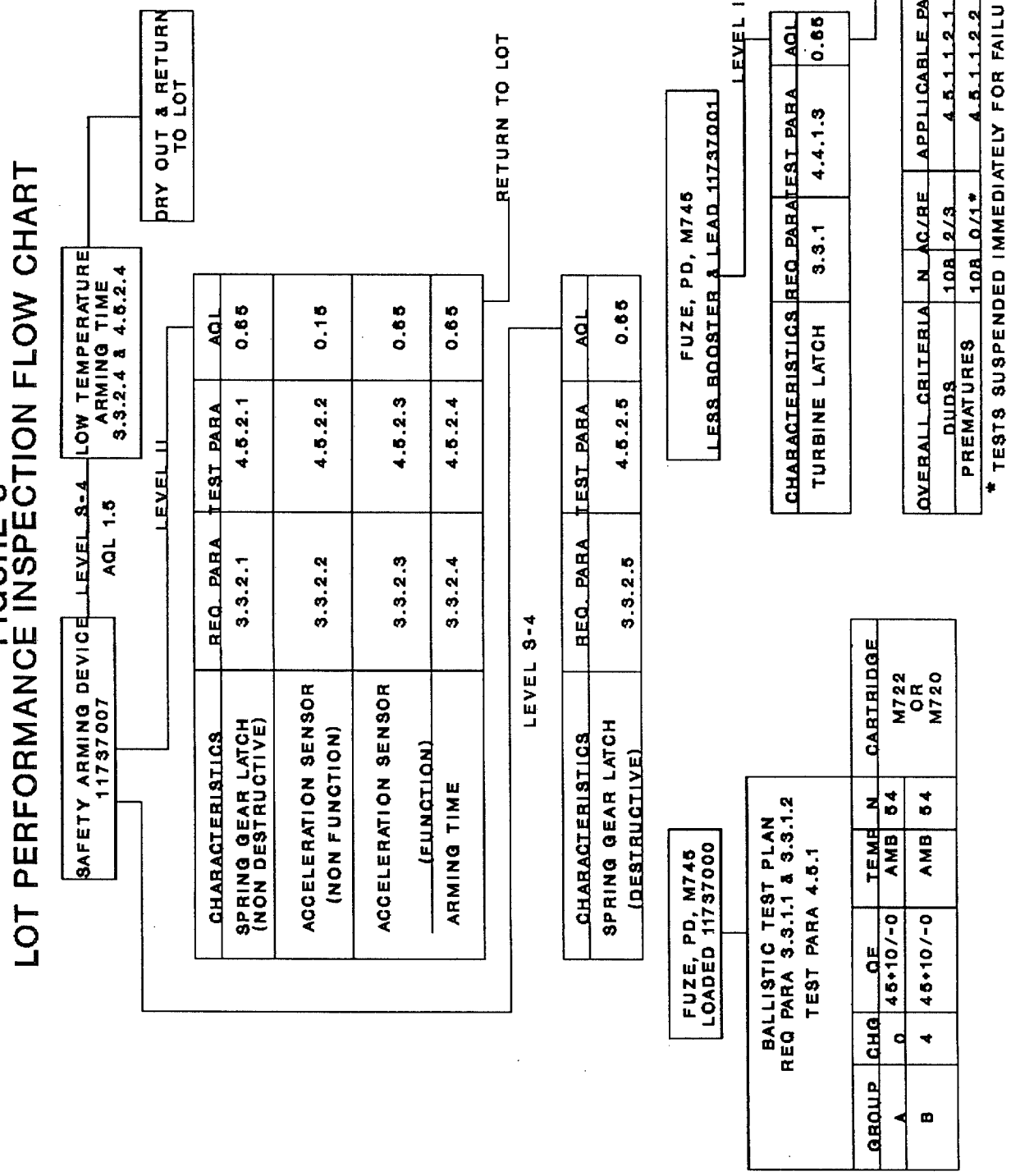
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# FIGURE 2
# INITIAL LOT PERFORMANCE INSPECTION FLOW CHART



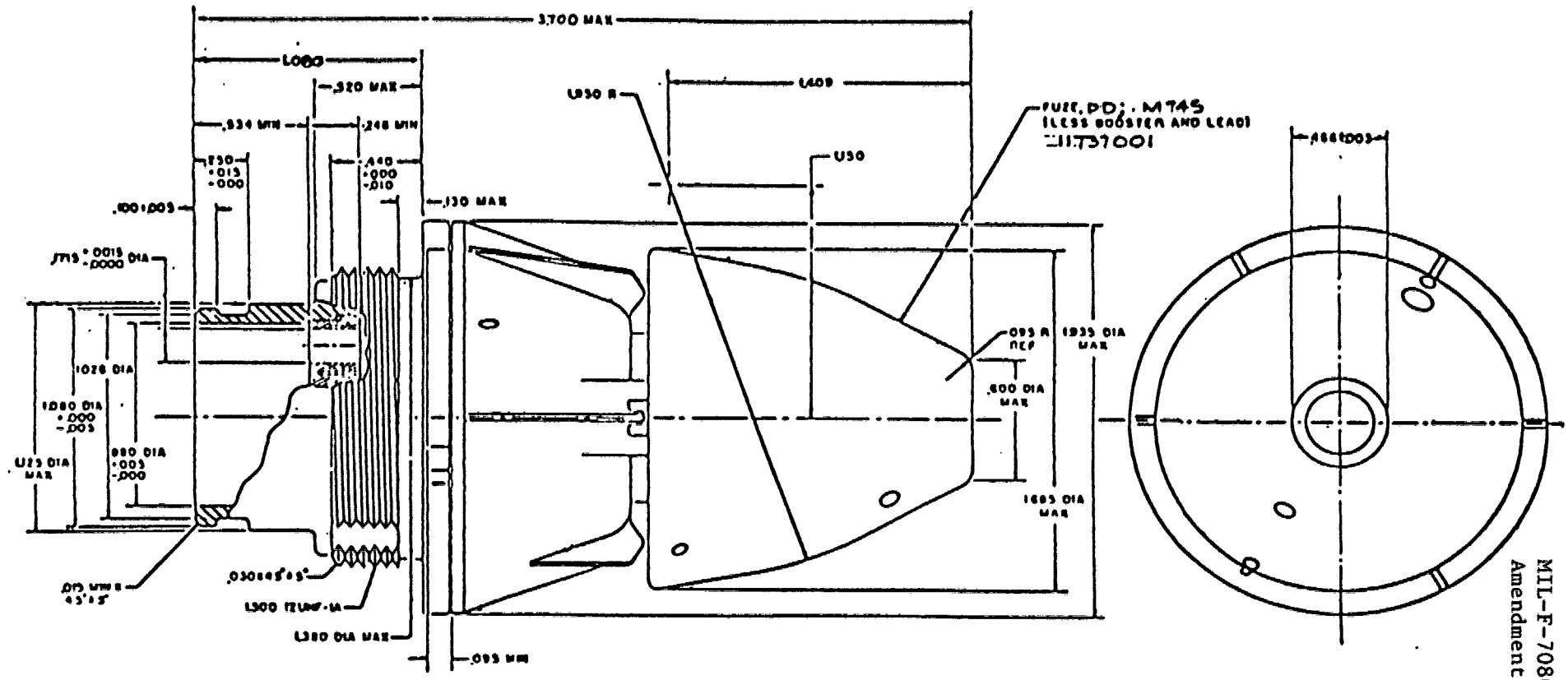

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# FIGURE 3
# LOT PERFORMANCE INSPECTION FLOW CHART






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NOTES

1. WEIGHT OF FULLY LOADED FUZE IS .50 + .00/-.10 LBS
2. .1715 DIA HOLE SHALL BE ADJACENT TO SCALLOP IN BOOSTER CAVITY SUCH THAT LEAD CAN BE INSERTED
3. S & A ROTOR SHALL BE IN SAFE POSITION (DWGS 11737010 AND 11737007) M61 DETONATOR (8663554) IN PRIMER HOUSING ASSY (11737005) SHALL NOT BE VISIBLE THROUGH BARRIER LEAD HOLE IN AFT END OF FUZE.
4. FUZE SHALL BE PERMANENTLY MARKED PER DWGS 11740819
5. FUZE SHOWN IS NOT FULLY LOADED SEE DWG NO 11737000 FOR LOADED ASSEMBLY AND COMPLETE INTERFACE DIMENSIONS

FIGURE 4 INTERFACE CONTROL DRAWING





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